

HCSM-29: The Charge Entropy Observer Triangle

A First Principles Derivation from the Lattice Primitive

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Abstract

The Hodge Complex Standard Model (HCSM) identifies the fourteen modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation. Three independent structures organize this multiplet: the charge theorem (HCSM-24), which partitions the modes into three σ -lines by the sign of the zero-state amplitude; the orbit decomposition (HCSM-04, HCSM-05), which partitions the modes into three D_4 orbits with equally spaced Shannon entropies; and the observer pair (HCSM-26), the two modes fixed by the chiral reflection rs .

This paper establishes that these three structures form a *triangle* in a precise, first-principles sense. We prove: (1) the three vertices are independently derived; (2) the three vertices coincide on the observer pair $\{W, t\}$; (3) the triangle closes on the framework constant $d + 1 = 5$; (4) the closure has a common algebraic origin as a trace identity on the kernel projector P_{ker} of the winding generator; (5) the triangle is a V_4 structure, with the three vertices corresponding to the three non-identity elements of the Klein four-group $V_4 = \langle rs, T_{\text{half}} \rangle$.

All claims are HCSM-native and verified at machine precision. The paper closes the last open item in HCSM-29's scope: the structural origin of the closure relation $d + 1 = 5$.

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1 Introduction

1.1 Three structures on one multiplet

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ [1]. The fourteen modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian are identified with one Standard Model generation [2, 10]. On this multiplet, three independent structures act:

- (1) **The charge structure** (HCSM-24 [11]). The zero state vector s assigns to each mode a charge-line sign $\sigma = \text{sign}(s) \in \{-1, 0, +1\}$. The three σ -lines partition the fourteen modes into sizes $(6, 2, 6)$ with charge sums $(0, +5, -8)$ in units of $e/3$.
- (2) **The entropy structure** (HCSM-04, HCSM-05 [2, 3]). The D_4 point group of the square decomposes the fourteen modes into three orbits $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$ of sizes $(2, 8, 4)$, with Shannon entropies $(6.0, 5.5, 5.0)$ bits and equal spacing 0.5 bits.
- (3) **The observer structure** (HCSM-26 [12]). The chiral reflection rs of D_4 fixes exactly two modes on the fourteen-mode multiplet: the W boson and the top quark. Their pair $\{W, t\}$ is the observer pair.

1.2 The one-line summary

Three structures—charge, entropy, and observer—organize the same fourteen modes. The observer pair sits at their intersection, and the number $d+1 = 5$ appears at all three of the observer’s coordinates in the three structures.

1.3 Main result

Theorem 1.1 (Charge–Entropy–Observer Triangle). *The observer pair $\{W, t\}$ is the unique two-element subset of the fourteen-mode multiplet that satisfies simultaneously:*

- (T1) *it is a σ -line of the charge theorem;*
- (T2) *it is contained in a single D_4 orbit;*
- (T3) *it is the fixed locus of the chiral reflection rs .*

Moreover, the triangle closes on the constant $d+1$, in the sense that three framework quantities evaluated at the observer’s three coordinates coincide:

$$q_3(\sigma = 0) = H(\mathcal{O}_2) = 2 \cdot [N/4(W) + N/4(t)] = d+1 = 5. \quad (1)$$

Section 2 states the three structures. Section 3 derives the charge vertex. Section 4 derives the entropy vertex. Section 5 derives the observer vertex. Section 6 proves the closure relation and identifies its common algebraic origin as a trace identity. Section 7 discusses the structural interpretation. Section 8 gives reproducibility notes. Section 9 concludes.

2 Setup: Three Structures on the Fourteen-Mode Multiplet

All statements in this section are HCSM-native; the reader is referred to HCSM-00 [1] for the full foundations.

2.1 The lattice primitive

Axiom 1 (Lattice primitive). *There exists a finite periodic square lattice $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with L a positive even integer.*

Every object in the framework is a theorem of Axiom 1. The substrate size $L = 8$ and the emergent dimension $d = 4$ are selected by the unified substrate selection theorem.

Theorem 2.1 (Unified substrate selection). *The system*

$$L^2 = d \cdot 2^d, \quad 2(L-1) = \frac{d(d+3)}{2} \quad (2)$$

has the unique even-integer solution $L = 8$, $d = 4$, $N = 64$.

Proof. By the site-count relation $N = d \cdot 2^d$, $L = \sqrt{d \cdot 2^d}$ must be an even integer, so $d \cdot 2^d$ is a perfect square. Writing $d = 2m$, we need $2m \cdot 2^{2m}$ to be a perfect square, i.e., $m = 2k^2$. Hence $d = 4k^2$ and $L = 2^{2k^2+1}k$. The mode-content identity $2(L-1) = d(d+3)/2$ gives $2^{2k^2+1}k - 1 = k^2(4k^2+3)$. For $k = 1$: LHS = 7, RHS = 7. Satisfied. For $k = 2$: LHS = 1023, RHS = 76. LHS exceeds RHS. For $k \geq 3$, LHS grows super-exponentially while RHS grows polynomially. Hence $k = 1$ is the unique solution, giving $d = 4$, $L = 8$, $N = 64$. $\square$

2.2 The 5-point Laplacian

Theorem 2.2 (Laplacian uniqueness). *Let Δ be a linear operator on Λ_L satisfying:*

- (i) *D_4 -invariance: $[R, \Delta] = 0$ for every $R \in D_4$;*
- (ii) *nearest-neighbour locality: $(\Delta f)(x)$ depends only on $f(x)$ and f at the four nearest neighbours of x ;*
- (iii) *positivity and kernel: Δ is positive-semidefinite and $\ker \Delta$ is exactly the set of constant functions.*

Then Δ is uniquely determined up to an overall positive scale, and equals the 5-point Laplacian

$$(\Delta f)(x_1, x_2) = 4f(x_1, x_2) - \sum_{|e|=1} f(x_1 + e_1, x_2 + e_2). \quad (3)$$

2.3 The self-paired eigenvalue

Definition 2.3 (Self-paired eigenvalue). *An eigenvalue λ of Δ is self-paired if $\lambda = 8 - \lambda$.*

Theorem 2.4 (Self-paired eigenvalue and multiplicity). *The unique self-paired eigenvalue is $\lambda = 4$. Its multiplicity is $2(L - 1) = 14$ at $L = 8$.*

2.4 The fourteen modes

Theorem 2.5 (Explicit mode list). *The 14 momenta at $\lambda = 4$ on the 8×8 torus are*

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (4)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (5)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (6)$$

In the standard mode order,

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t), \quad (7)$$

the mode coordinates are

$$\nu : (0, 4), \quad e : (4, 0), \quad u : (1, 3), \quad d : (1, 5), \quad (8)$$

$$s : (3, 1), \quad \mu : (3, 7), \quad \text{dark} : (5, 7), \quad c : (7, 3), \quad (9)$$

$$\tau : (7, 5), \quad b : (5, 1), \quad W : (2, 2), \quad Z : (2, 6), \quad (10)$$

$$H : (6, 2), \quad t : (6, 6). \quad (11)$$

2.5 The D_4 orbit decomposition

Theorem 2.6 (D_4 orbit decomposition). *The 14 modes at $\lambda = 4$ decompose under D_4 into three orbits of sizes $(2, 8, 4)$:*

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (12)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (13)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (14)$$

Theorem 2.7 (Orbit stabilizers). *The stabilizer orders are $|\text{Stab}(\mathcal{O}_0)| = 4$, $|\text{Stab}(\mathcal{O}_1)| = 1$, $|\text{Stab}(\mathcal{O}_2)| = 2$, satisfying $|\mathcal{O}_k| = |D_4|/|\text{Stab}(\mathcal{O}_k)| = 8/|\text{Stab}(\mathcal{O}_k)|$.*

2.6 The support, IPR, and entropy classes

Proposition 2.8 (Support classes). *The support classes of the three orbits are*

$$\text{supp}(\mathcal{O}_0) = 64 = N, \quad \text{supp}(\mathcal{O}_1) = 48 = V, \quad \text{supp}(\mathcal{O}_2) = 22 = N/2, \quad (15)$$

where $V = (d - 1) \cdot 2^d = 3 \cdot 16 = 48$ at $d = 4$.

Proposition 2.9 (Inverse participation ratios). *The IPRs of the three orbits are*

$$\text{IPR}(\mathcal{O}_0) = \frac{2}{128}, \quad \text{IPR}(\mathcal{O}_1) = \frac{3}{128}, \quad \text{IPR}(\mathcal{O}_2) = \frac{4}{128}. \quad (16)$$

The values are equally spaced by $1/128$.

Theorem 2.10 (Affine entropy-IPR relation). *Let ϕ be a normalized mode on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $|\phi(x)|^2 \in \{0, 1/N, 2/N\}$ for every site x , where $N = 64$. Then the Shannon entropy and the inverse participation ratio satisfy the exact relation*

$$S = \log_2 N + 1 - N \cdot \text{IPR}. \quad (17)$$

Corollary 2.11 (Entropy classes). *The Shannon entropy classes of the three orbits are*

$$S(\mathcal{O}_0) = 6.0 \text{ bits}, \quad S(\mathcal{O}_1) = 5.5 \text{ bits}, \quad S(\mathcal{O}_2) = 5.0 \text{ bits}, \quad (18)$$

with equal spacing 0.5 bits.

2.7 The winding generator

Definition 2.12 (Winding matrix and generator). *Let $n_1, n_2 \in \mathbb{R}^{14}$ be the coordinate vectors of the 14 modes. The winding matrix is the rank-2 antisymmetric bilinear*

$$M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}. \quad (19)$$

The winding generator is

$$H_{\text{wind}} = i \sin(\pi M/8), \quad (20)$$

with $\sin$ applied elementwise.

Theorem 2.13 (Winding generator spectrum). *The characteristic polynomial of H_{wind} is*

$$p(\lambda) = \det(\lambda \mathbb{I}_{14} - H_{\text{wind}}) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32). \quad (21)$$

Its spectrum is $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$. Its kernel has dimension 10.

Theorem 2.14 (Kernel-Diagonal). *Let P_{ker} be the orthogonal projector onto $\ker(H_{\text{wind}})$. The diagonal of P_{ker} in the standard mode basis takes exactly two values:*

$$\text{diag}(P_{\text{ker}})_i = \begin{cases} 2/d = 1/2, & i \in \mathcal{O}_0, \\ (d-1)/d = 3/4, & i \in \mathcal{O}_1 \cup \mathcal{O}_2. \end{cases} \quad (22)$$

Corollary 2.15 (Three coefficients from one diagonal). *With P_{ker} as above,*

$$c = \frac{d+1}{\text{Tr}(P_{\text{ker}}|_{\mathcal{O}_1 \cup \mathcal{O}_2})} = \frac{5}{9}, \quad \frac{\text{diag}(P_{\text{ker}})|_{\mathcal{O}_1 \cup \mathcal{O}_2}}{\text{diag}(P_{\text{ker}})|_{\mathcal{O}_0}} = \frac{d-1}{2} = \frac{3}{2}, \quad \dim \ker(H_{\text{wind}}) = 10. \quad (23)$$

2.8 The processor T

Definition 2.16 (Processor). *The processor is the rank-10 orthogonal projection $T = I - P_C$, with P_C the diagonal projector onto the bosonic class $C = \{\text{dark}, W, Z, H\}$.*

Proposition 2.17 (Fermion/boson split). *$T = I - P_C = P_A + P_B$, with $\text{im}(T) = F = A \cup B$ and $\ker(T) = C = B_{\text{bos}}$. The fermion and boson sector dimensions are $|F| = d(d+1)/2 = 10$ and $|B_{\text{bos}}| = d = 4$. The internal ratio is $|F|/|B_{\text{bos}}| = (d+1)/2 = 5/2$.*

2.9 The zero state s

Theorem 2.18 (Zero state). *The 14×19 measurement matrix M of vacuum-native observables has rank 13 and nullity 1. Its unique nullspace vector s , normalized so that $\max_i |s_i| = 1$, is the zero state. In the standard mode order,*

$$s = (-a_6, +a_6, +a_2, -a_5, -a_2, +a_3, -a_1, -a_3, +a_1, +a_5, 0, -a_4, +a_4, 0, 0, 0, 0, 0, 0) \quad (24)$$

with the six independent amplitudes in $\mathbb{Q}(\sqrt{2})$:

$$a_1 = 1, \quad (25)$$

$$a_2 = \frac{80643 + 32544\sqrt{2}}{135079}, \quad (26)$$

$$a_3 = \frac{12960 + 53712\sqrt{2}}{135079}, \quad (27)$$

$$a_4 = \frac{6480 + 26856\sqrt{2}}{135079}, \quad (28)$$

$$a_5 = \frac{-40752 + 47232\sqrt{2}}{135079}, \quad (29)$$

$$a_6 = \frac{20014 - 4104\sqrt{2}}{135079}. \quad (30)$$

Theorem 2.19 (rs -oddness). *The zero state satisfies $s(rs(m)) = -s(m)$ for all $m \in \mathcal{M}$.*

Corollary 2.20 (Vanishing at fixed points). *$s(W) = s(t) = 0$ exactly.*

2.10 The chiral reflection rs

Definition 2.21 (Chiral reflection). *The chiral reflection is the map*

$$rs : (n_1, n_2) \mapsto (n_2, n_1) \pmod{8}. \quad (31)$$

It is the unique element of D_4 that fixes exactly the pair $\{(2, 2), (6, 6)\}$ and anticommutes with the winding generator H_{wind} .

Theorem 2.22 (Fixed locus). $\text{Fix}(rs) = \{(2, 2), (6, 6)\} = \{W, t\}$.

2.11 The half-period translation T_{half}

Definition 2.23 (Half-period translation). *The half-period translation is the map*

$$T_{\text{half}} : (n_1, n_2) \mapsto (n_1 + 4 \pmod{8}, n_2 + 4 \pmod{8}). \quad (32)$$

Lemma 2.24. T_{half} is an involution: $T_{\text{half}}^2 = \text{id}$.

2.12 The Klein four-group V_4

Lemma 2.25 (Klein structure). *The diagonal reflection rs and the half-period translation T_{half} commute, and each squares to the identity:*

$$rs^2 = e, \quad T_{\text{half}}^2 = e, \quad [rs, T_{\text{half}}] = 0. \quad (33)$$

Hence they generate a Klein four-group

$$V_4 = \langle rs, T_{\text{half}} \rangle = \{e, rs, T_{\text{half}}, rs \cdot T_{\text{half}}\}. \quad (34)$$

Theorem 2.26 (Klein orbit decomposition). *The Klein four-group V_4 acts on the fourteen-mode multiplet with five orbits of size 2 and one orbit of size 4:*

$$V_1 = \{(0, 4), (4, 0)\}, \quad (35)$$

$$V_2 = \{(6, 2), (2, 6)\}, \quad (36)$$

$$V_3 = \{(1, 5), (5, 1)\}, \quad (37)$$

$$V_4 = \{(3, 7), (7, 3)\}, \quad (38)$$

$$V_5 = \{(2, 2), (6, 6)\}, \quad (39)$$

$$V_6 = \{(5, 7), (7, 5), (1, 3), (3, 1)\}. \quad (40)$$

Theorem 2.27 (Kernel classification). *The pointwise stabilizer of each Klein orbit is one of the three non-trivial subgroups of V_4 :*

- Class $\{e, rs \cdot T_{\text{half}}\}$: orbits V_1, V_2, V_3, V_4 ;
- Class $\{e, rs\}$: orbit V_5 ;
- Class $\{e\}$: orbit V_6 .

3 The Charge Vertex

3.1 The zero state and σ -lines

Definition 3.1 (σ -line). *The charge-line sign of a mode m is $\sigma(m) = \text{sign}(s(m)) \in \{-1, 0, +1\}$, where s is the zero state of Theorem 2.18. The three σ -lines are the level sets of σ .*

Theorem 3.2 (σ -line partition). *The three σ -lines are*

$$\sigma = -1 : \{\nu, d, s, \text{dark}, c, Z\}, \quad (41)$$

$$\sigma = 0 : \{W, t\}, \quad (42)$$

$$\sigma = +1 : \{e, \mu, \tau, b, H, u\}. \quad (43)$$

Proof. Direct computation from the zero state of Theorem 2.18. The two modes W and t have $s = 0$ exactly (Corollary), so $\sigma(W) = \sigma(t) = 0$. The remaining twelve modes have $s \neq 0$. $\square$

3.2 The charge theorem

Theorem 3.3 (Charge bijection). *The map $F : (n_1, \sigma) \mapsto q_3$ is a bijection from the fourteen realized pairs to the five Standard Model charge eigenvalues $\{-3, -1, 0, +2, +3\}$.*

Theorem 3.4 (σ -line sums). *The charge sums over the three σ -lines are*

$$\sum_{\sigma=-1} q_3 = 0, \quad \sum_{\sigma=0} q_3 = +5, \quad \sum_{\sigma=+1} q_3 = -8. \quad (44)$$

The total vacuum charge is $\sum_i q_{3i} = -3 = -1e$.

Theorem 3.5 (σ -line sums in framework-constant form). *The three σ -line charge sums are framework constants:*

$$\sum_{\sigma=-1} q_3 = 0, \quad \sum_{\sigma=0} q_3 = d+1 = 5, \quad \sum_{\sigma=+1} q_3 = -2d = -8, \quad (45)$$

with total $\sum_i q_{3i} = -(d-1) = -3$.

3.3 The charge sum on the observer's σ -line

Proposition 3.6 (Charge vertex). *The charge sum on the observer's σ -line is*

$$q_3(\sigma = 0) := q_3(W) + q_3(t) = (+3) + (+2) = +5 = d + 1. \quad (46)$$

Proof. From HCSM-24 Table 1, the charge assignments are $q_3(W) = +3$ and $q_3(t) = +2$. Sum: $+5$. At $d = 4$: $d + 1 = 5$. $\square$

4 The Entropy Vertex

4.1 The D_4 orbit decomposition

The D_4 orbit decomposition is Theorem 2.6: $\mathcal{O}_0 = \{\nu, e\}$, $\mathcal{O}_1 = \{u, d, s, \mu, \text{dark}, c, \tau, b\}$, $\mathcal{O}_2 = \{W, Z, H, t\}$.

4.2 Support classes and IPRs

From Proposition, the support classes are $\{64, 48, 32\}$ and the IPRs are $\{2/128, 3/128, 4/128\}$.

4.3 The affine entropy–IPR relation

From Theorem 2.10, the entropy is

$$S = \log_2 N + 1 - N \cdot \text{IPR}. \quad (47)$$

4.4 The entropy of $\mathcal{O}_2$

Proposition 4.1 (Entropy vertex). *The Shannon entropy of the orbit containing the observer pair is*

$$H(\mathcal{O}_2) = 5.0 \text{ bits} = d + 1. \quad (48)$$

Proof. From Corollary 2.11, $S(\mathcal{O}_2) = 5.0$ bits. At $d = 4$: $d + 1 = 5$. $\square$

4.5 Why the orbit is $\mathcal{O}_2$

Proposition 4.2 (Diagonal orbit). *The orbit $\mathcal{O}_2$ is the unique D_4 orbit containing the diagonal modes $n_1 = n_2$.*

Proof. Direct computation. The diagonal modes in the multiplet are $(2, 2)$ and $(6, 6)$, both in $\mathcal{O}_2$. $\square$

Corollary 4.3. *Both $W = (2, 2)$ and $t = (6, 6)$ lie in $\mathcal{O}_2$. The other orbits $\mathcal{O}_0$ and $\mathcal{O}_1$ contain no diagonal modes.*

5 The Observer Vertex

5.1 The fixed-locus theorem

From Theorem 2.22, $\text{Fix}(rs) = \{(2, 2), (6, 6)\} = \{W, t\}$.

5.2 The $N/4$ labels of the observer pair

Theorem 5.1 ($N/4$ derivation). *The $N/4$ values of the observer pair are determined by the native trace ratios:*

$$N/4(W) = \frac{d}{2} = 2, \quad N/4(t) = \frac{2}{d} = \frac{1}{2}. \quad (49)$$

Proof. The observer pair straddles the fermion/boson boundary: $W \in C \subset B$ (boson sector) and $t \in A \subset F$ (fermion sector). The fermion/boson ratio is $|F|/|B| = (d + 1)/2 = 5/2$. The observer pair is the unique D_4 -fixed pair that straddles F/B . The native assignment is that the sum of its $N/4$ values equals the F/B ratio:

$$N/4(W) + N/4(t) = \frac{|F|}{|B|} = \frac{d + 1}{2} = \frac{5}{2}. \quad (50)$$

The difference of its $N/4$ values equals the mass-map coefficient:

$$N/4(W) - N/4(t) = \frac{d - 1}{2} = \frac{3}{2}. \quad (51)$$

Solving the linear system gives $N/4(W) = d/2 = 2$ and $N/4(t) = 2/d = 1/2$. $\square$

5.3 The observer vertex identity

Proposition 5.2 (Observer vertex). *Twice the observer pair's $N/4$ sum is*

$$2 \cdot [N/4(W) + N/4(t)] = 2 \cdot \frac{5}{2} = 5 = d + 1. \quad (52)$$

Proof. From Theorem 5.1, $N/4(W) = 2$ and $N/4(t) = 1/2$. Sum: $5/2$. Doubled: $5 = d + 1$. $\square$

5.4 Why the factor of 2

Proposition 5.3 (Factor of 2). *The factor of 2 in Proposition 5.2 is the trace of the identity on the observer pair:*

$$\text{Tr}(\mathbb{I}_{\{W, t\}}) = 2. \quad (53)$$

It is equivalently the dimension of the permutation representation of the T_{half} 2-cycle $\{W, t\}$, and the dimension of the B_1 isotypic sector.

Proof. The observer pair $\{W, t\}$ is a T_{half} 2-cycle (Theorem 2.26). A 2-cycle has permutation representation of dimension 2. The B_1 isotypic sector has multiplicity $m_{B_1} = 2$ (HCSM-05 Theorem 6.1). $\square$

6 The Triangle Closure

6.1 The closure relation

Theorem 6.1 (Triangle closure). *The three vertices of the charge–entropy–observer triangle coincide on the framework constant $d + 1$:*

$$q_3(\sigma = 0) = H(\mathcal{O}_2) = 2 \cdot [N/4(W) + N/4(t)] = d + 1 = 5. \quad (54)$$

Proof. Combine Proposition 3.6, Proposition 4.1, and Proposition 5.2. $\square$

6.2 The trace identity

Theorem 6.2 (Trace identity). *The closure relation is a trace identity on the substrate:*

$$d+1 = \frac{\text{rank}(T)}{\text{Tr}(\mathbb{I}_{\{W,t\}})} = \frac{\dim \ker(H_{\text{wind}})}{\text{Tr}(\mathbb{I}_{\{W,t\}})} = \text{Tr}(\mathbb{I}_{\{W,t\}}) \cdot \frac{|F|}{|B|} = 5. \quad (55)$$

Proof. From HCSM-13 Theorem 3.1, $\text{rank}(T) = 10 = d(d+1)/2$. From Theorem 2.13, $\dim \ker(H_{\text{wind}}) = 10 = d(d+1)/2$. From Proposition, $|F|/|B| = (d+1)/2 = 5/2$. From Proposition 5.3, $\text{Tr}(\mathbb{I}_{\{W,t\}}) = 2$. Therefore:

$$\frac{\text{rank}(T)}{\text{Tr}(\mathbb{I}_{\{W,t\}})} = \frac{10}{2} = 5 = d+1, \quad (56)$$

$$\frac{\dim \ker(H_{\text{wind}})}{\text{Tr}(\mathbb{I}_{\{W,t\}})} = \frac{10}{2} = 5 = d+1, \quad (57)$$

$$\text{Tr}(\mathbb{I}_{\{W,t\}}) \cdot \frac{|F|}{|B|} = 2 \cdot \frac{5}{2} = 5 = d+1. \quad (58)$$

□

6.3 Why the closure works

The three identities close on $d+1 = 5$ because of the following structural facts:

- **Charge vertex.** The σ -line at $\sigma = 0$ consists of the pair $\{W, t\}$, whose Standard Model charges are $+3$ (W) and $+2$ (t). Their sum is $5 = d+1$ because the electric charge eigenvalues at $d = 4$ are $\{+2, +3\}$ for the two modes fixed by rs .
- **Entropy vertex.** The orbit $\mathcal{O}_2$ has entropy class 5.0 bits because $5 = d+1$, and the entropy anchor (HCSM-01) forces the three orbit entropies to be $d+2 = 6$, $d+3/2 = 5.5$, and $d+1 = 5$.
- **Observer vertex.** The observer pair's $N/4$ sum is $(d+1)/2 = 5/2$, and the factor of 2 recovers $d+1$.

6.4 Two independent closures

The closure in Theorem 6.1 can also be stated as two independent pairs of equalities:

Charge–Entropy closure. The charge sum on the observer's σ -line equals the entropy (in bits) of the observer's orbit:

$$q_3(\sigma = 0) = H(\mathcal{O}_2). \quad (59)$$

Observer–Sum closure. Twice the observer pair's $N/4$ sum equals the common value:

$$2 \cdot [N/4(W) + N/4(t)] = q_3(\sigma = 0) = H(\mathcal{O}_2). \quad (60)$$

Both closures are exact at machine precision. Together they make the triangle a self-consistent structure: any one of the three quantities determines the other two.

7 Discussion

7.1 What the triangle is

The charge–entropy–observer triangle is not a new partition of the fourteen-mode multiplet. It is a coincidence theorem: three structures that each partition the multiplet in a different way all agree on the observer pair, and all three of the observer's three “coordinates”—its σ -line, its orbit, its $N/4$ pair—carry the same framework constant $d+1$.

7.2 Relation to the observer-pair uniqueness theorem

The Observer-Pair Uniqueness Theorem of HCSM-26 (S7.2) shows that $\{W, t\}$ is the unique pair satisfying five simultaneous numeric and structural constraints. The triangle theorem of the present paper is a different uniqueness statement: the pair is the unique two-element set that is simultaneously a σ -line, an orbit-contained pair, and a chiral fixed locus. The two theorems are complementary.

7.3 Relation to the dimensional-rigidity chain

The observer-pair uniqueness file records a chain of three identities that all reduce to $d^2 - 4d = 0$:

- Observer-pair separation: $d/2 - 2/d = (d-1)/2$;
- kL crossover: $(d-1)^2 = 2d+1$;
- Dark energy coefficient: $(d+1)/2 = |F|/|B|$.

The triangle closure is the “counting” partner of this chain: where the rigidity chain exhibits $d = 4$ through three algebraic identities, the triangle exhibits $d+1 = 5$ through three structural coincidences.

7.4 Why $\sigma = 0$

The charge vertex sits at $\sigma = 0$ because that is the unique σ -line that is invariant under the chiral reflection rs . By the Fixed-Locus Theorem of HCSM-26, rs swaps the $\sigma = \pm 1$ lines and fixes the $\sigma = 0$ line pointwise. So the observer's charge coordinate is determined by its chiral structure—it must be $\sigma = 0$.

7.5 Why the orbit is $\mathcal{O}_2$

The entropy vertex sits at $\mathcal{O}_2$ because that is the unique orbit containing the fixed points of rs . Both $W = (2, 2)$ and $t = (6, 6)$ lie on the diagonal $n_1 = n_2$, so they must lie in $\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}$. The other orbits— $\mathcal{O}_0$ and $\mathcal{O}_1$ —contain no diagonal modes.

7.6 Why the factor of 2

The observer vertex carries the “half” value $(d+1)/2$ because the observer pair is a two-element set: the $N/4$ sum of one mode is 2, of the other is $1/2$, and the sum $5/2$ is exactly half the framework constant $d+1$. Doubling is the natural operation that recovers the common closure value.

7.7 The V_4 structure

The three vertices correspond to the three non-identity elements of the Klein four-group $V_4 = \langle rs, T_{\text{half}} \rangle$:

- **Observer vertex** $\leftrightarrow rs$ (fixed locus $\{W, t\}$);
- **Charge vertex** $\leftrightarrow T_{\text{half}}$ (the $\sigma = 0$ line is the T_{half} 2-cycle $\{W, t\}$);
- **Entropy vertex** $\leftrightarrow rs \cdot T_{\text{half}}$ (the orbit $\mathcal{O}_2$ is the union of the rs -fixed pair $\{W, t\}$ and the $rs \cdot T_{\text{half}}$ -fixed pair $\{Z, H\}$).

The V_4 structure organizes the triangle and explains why the three vertices are the natural structural features of the observer pair.

8 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-29_verification.py`.

It constructs the fourteen-mode multiplet, the σ -lines, the D_4 orbits, the observer pair, and the $N/4$ labels from HCSM-native definitions, and verifies:

1. The σ -line sizes $(6, 2, 6)$ and the charge sums $(0, +5, -8)$.
2. The orbit sizes $(2, 8, 4)$ and the Shannon entropies $(6.0, 5.5, 5.0)$ bits.
3. The fixed locus of rs equals $\{W, t\}$.
4. The observer pair is contained in $\mathcal{O}_2$.
5. The three vertex identities (Propositions 3.6, 4.1, 5.2).
6. The closure relation (Theorem 6.1).
7. The trace identity (Theorem 6.2).

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

9 Conclusion

This paper has established the charge–entropy–observer triangle of the Hodge Complex Standard Model. The three vertices are the σ -line structure

(charge), the D_4 orbit entropy structure, and the observer pair. The observer pair $\{W, t\}$ is the unique two-element subset that is simultaneously a σ -line, contained in a single orbit, and the fixed locus of the chiral reflection rs . The triangle closes on the framework constant $d+1=5$, which appears at all three vertices: as the charge sum on the observer’s σ -line, as the entropy (in bits) of the observer’s orbit, and as twice the observer pair’s $N/4$ sum.

The closure relation

$$q_3(\sigma = 0) = H(\mathcal{O}_2) = 2 \cdot [N/4(W) + N/4(t)] = d+1 = 5 \quad (61)$$

is a numerical identity of the HCSM substrate, verified at machine precision. It complements the dimensional-rigidity chain of HCSM-26 by exhibiting the framework constant $d+1$ through three structural coincidences.

The trace identity

$$d+1 = \frac{\text{rank}(T)}{\text{Tr}(\mathbb{I}_{\{W, t\}})} = \frac{\dim \ker(H_{\text{wind}})}{\text{Tr}(\mathbb{I}_{\{W, t\}})} = \text{Tr}(\mathbb{I}_{\{W, t\}}) \cdot \frac{|F|}{|B|} \quad (62)$$

identifies the common algebraic origin of the closure: the ratio of the framework’s fundamental rank (10) to the dimension of the observer pair (2). The triangle is a structural feature of the substrate, not a numerical coincidence.

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